Expected values

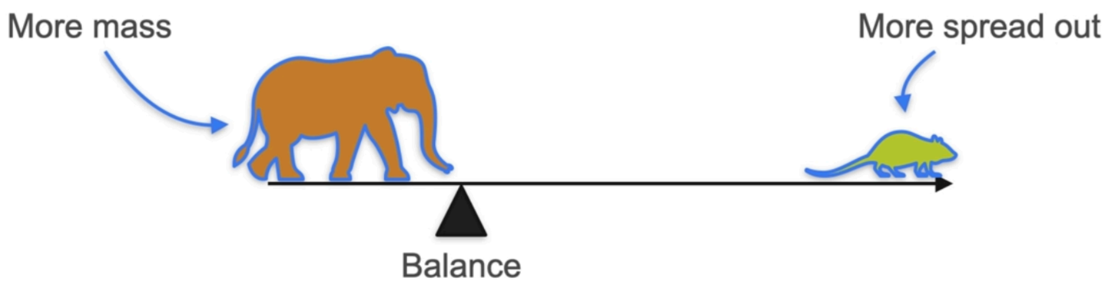
Expected values can be used to describe some interesting properties of variance as well as some advanced properties of probability distributions, such as skewness and kurtosis.

Discrete random variables	Continuous random variables
$\mathbb{E}[X] = \sum_{x} x \, p_X(x)$	$\int_{-\infty}^{\infty} x \cdot f_X(x) \, dx$
Weighted using PMF	Weighted using PDF

So basically, the expected value is the mean

The expected value $E(X)$ is mathematically equivalent to the theoretical mean.

The mean is not always in the middle,

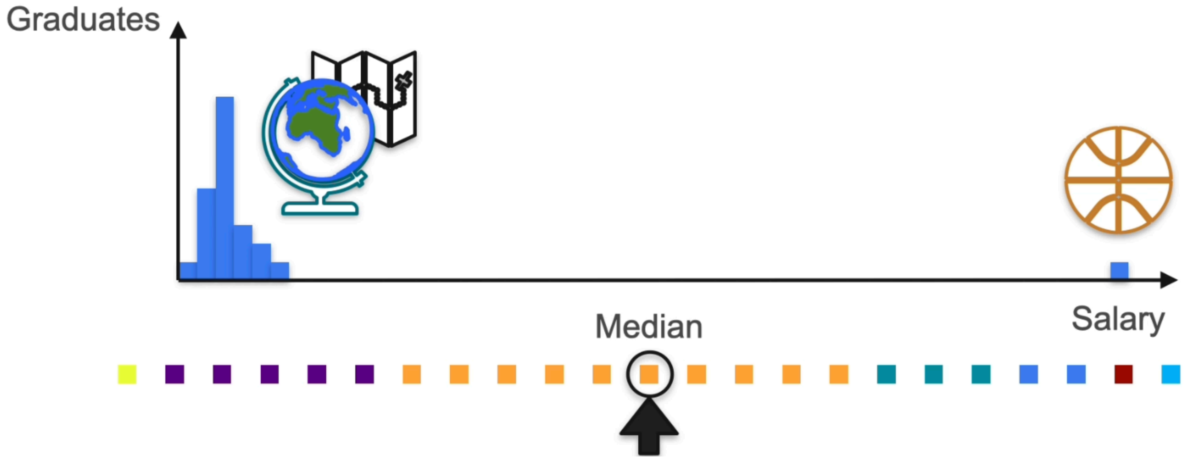


but the median should be always in the middle!

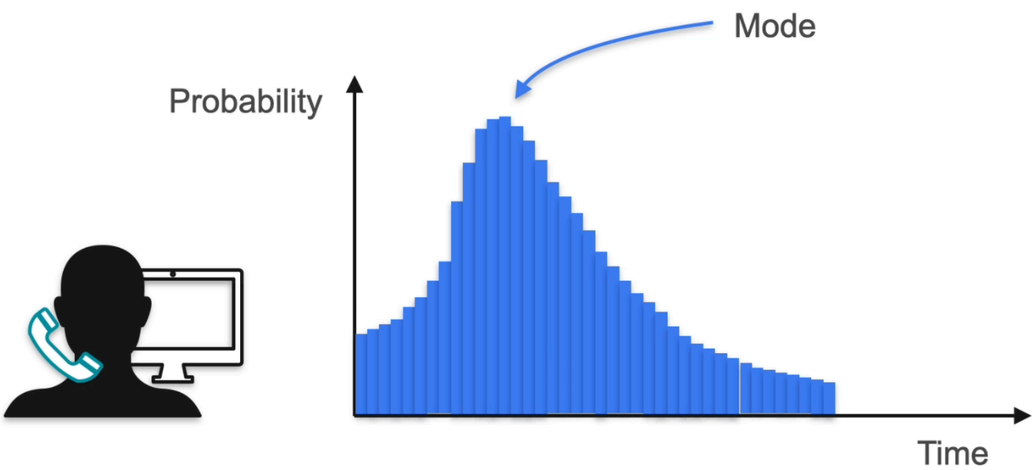
But the mean is not alway reliable because there is OUTLIER in real world.

and to solve that problem

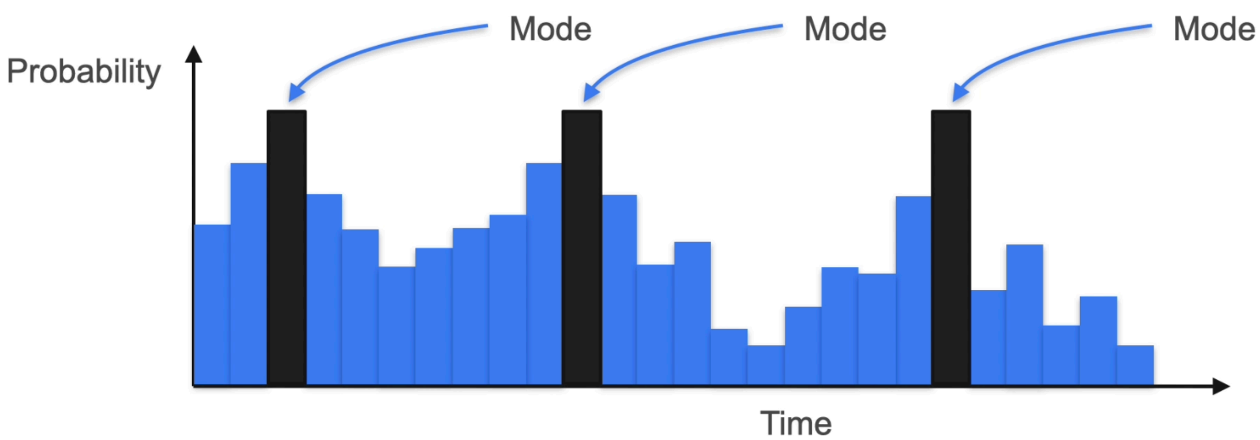
we list those data in order, and then calculate the median



Mode = it is the value at which the distribution has the highest probability, which makes it the most frequent one.



Mode = The value that has the highest height



In the normal distribution, the mode, mean, and median have the same value!

Sum of Expectation

$$E[X +] = E[X] + E[]$$

The expected value = is the average outcome we'd expect if we repeated an experiment many times.

Variance

$$Variance = E[(X - E[X])^2]$$

Standard Deviation

it is a useful way to measure the spread of a distribution using the same units of the distributions.

The 68-95-99.7 rule in Normal Distribution